

Analysis I

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Lecture 1 – 22/01/26

Analysis is about things that are small, things that are large, and quantifyig how small or large they are.

§1 Numerical sequences and series

§1.1 Basics

Definition 1.1. A sequence $(x_n)_{n \in \mathbb{N}}$ on a set X is an enumerated list where each element is in X .

In this chapter, $X \subseteq \mathbb{R}$ or $X \subseteq \mathbb{C}$.

What does it mean for a sequence (x_n) to converge to a point $x \in X$? What does it mean for (x_n) to diverge to infinity?

For convergence, we want $|x_n - x|$ to be small. But what is small? In fact, we want $|x_n - x|$ to be arbitrarily small, i.e. smaller than any $\varepsilon > 0$ we choose. What really matters is the *tail* of the sequence – we can always ignore the first N terms, and we should pick N depending on ε .

For divergence to infinity, we want $|x_n|$ to get arbitrarily large, i.e. larger than any $L > 0$ we choose. Again, we ignore the first N terms depending on L .

Definition 1.2. We say (x_n) converges to some finite x , $x_n \rightarrow x$, if, for any $\varepsilon > 0$, there exists $N(\varepsilon)$ such that $|x_n - x| < \varepsilon$ for all $n \geq N$.

We say (x_n) diverges to infinity if, for any $L > 0$, there exists $N(L)$ such that $x_n > L$ for all $n \geq N$, analogously for negative infinity. For complex sequences, we must consider $|x_n|$.

- Example 1.3**
- Take $x_n = \frac{1}{n}$, then $x_n \rightarrow 0$ because $|x_n - 0| = \frac{1}{n} < \varepsilon$ for any $n \geq N = 1 + \lceil \frac{1}{\varepsilon} \rceil$.
 - Take $x_n = \frac{1}{2^n}$, then $x_n \rightarrow 0$ because $|x_n - 0| = \frac{1}{2^n} < \varepsilon$ for any $n \geq N = \max(1 + \lceil \log_2(\frac{1}{\varepsilon}) \rceil, 1)$.
 - Take $x_n = in$, then $x_n \rightarrow \infty$, since $\frac{x_n}{i} = n > L$ for any $n \geq N = 1 + \lceil L \rceil$.
 - Take $x_n = (-1)^n$, then x_n doesn't converge or diverge to infinity.

Lemma 1.4

If it exists, the limit of (x_n) is unique.

Proof. Suppose $x_n \rightarrow a$ and $x_n \rightarrow b$. Take $\varepsilon > 0$, then we have

$$\begin{aligned} \exists N_1 = N_1(\varepsilon) \text{ s.t. } |x_n - a| < \varepsilon \quad \forall n \geq N_1 \\ \exists N_2 = N_2(\varepsilon) \text{ s.t. } |x_n - b| < \varepsilon \quad \forall n \geq N_2 \end{aligned}$$

Now take $N = \max(N_1, N_2)$. We have

$$\begin{aligned} |a - b| &= |a - x_n + x_n - b| \\ &\leq |x_n - a| + |x_n - b| \\ &< 2\varepsilon \end{aligned}$$

for all $\varepsilon > 0$. It follows that $|a - b| = 0$, so $a = b$. ■

This justifies the notation $\lim_{n \rightarrow \infty} x_n = x$.

Proposition 1.5 (Sandwich theorems)

Let $(x_n), (y_n), (z_n)$ be real sequences. Then

- If $x_n \leq y$ for all n and $x_n \rightarrow x$, then $x \leq y$. Note that $x_n < y$ does not imply $x < y$.
- If $x_n \rightarrow x$, $z_n \rightarrow x$ and $x_n \leq y_n \leq z_n$ for all n , then $y_n \rightarrow x$.

Proof. Left as an exercise. ■

Lemma 1.6

Let (x_n) be a complex sequence. Then $x_n \rightarrow x$ if and only if $\operatorname{Re}(x_n) \rightarrow \operatorname{Re}(x)$ and $\operatorname{Im}(x_n) \rightarrow \operatorname{Im}(x)$.

Proof. Recall that, for $z \in \mathbb{C}$, $|z| = \sqrt{\operatorname{Re}(z)^2 + \operatorname{Im}(z)^2}$.

$\implies$ Use $|\operatorname{Re}(z)| \leq |z|$ and $|\operatorname{Im}(z)| \leq |z|$ with $z = x_n - x$.

$\Leftarrow$ Use $|z| \leq |\operatorname{Re}(z)| + |\operatorname{Im}(z)|$, again with $z = x_n - x$. Fixing $\varepsilon > 0$, we can get $|x_n - x| < 2\varepsilon$, so we're done. ■

Lemma 1.7

Let $x_n \rightarrow x$, $y_n \rightarrow y$. Then $x_n + y_n \rightarrow x + y$, $x_n y_n \rightarrow xy$.

Proof. The first part is analogous to the above result on complex sequences and is an exercise.

For the second part, we have

$$\begin{aligned} |x_n y_n - xy| &= |x_n y_n - x y_n + x y_n - xy| \\ &\leq |y_n| |x_n - x| + |x| |y_n - y| \end{aligned}$$

Now, for $\varepsilon > 0$, we can pick an N to make $|x_n - x| < \varepsilon$, $|y_n - y| < \varepsilon$. Hence

$$\begin{aligned} |x_n y_n - xy| &< \varepsilon(|x| + |y_n|) \\ &< \varepsilon(|x| + Y) \end{aligned} \quad \text{by lemma 1.8, } Y > 0$$
■

Lemma 1.8

Convergent sequences are bounded.